MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth Semester

Discipline Specific Core Course (DSC) - MG4DSCGEO202 - FOSSILOLOGY

ANSWER SCHEME

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

(Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap))

Part A: Short Answer (60 words)

Answer any 7. Each carries 2 marks.

1. Give any two examples of Fauna. [K] / [CO1]

- Any two specific animal groups or species (e.g., Trilobites, Brachiopods, Mammals). [1 Mark + 1 Mark]

2. Define umbo in brachiopods. [K] / [CO2]

- The arch part just behind the beak of each valve, located at the posterior end. [2 Mark]

3. What is the primary criterion for classifying plant fossils? [U] / [CO3]

- The **morphological features** of the preserved plant parts (leaves, stems, seeds, or spores). [2 Mark]

4. Define index fossil. [U] / [CO1]

- A fossil used for dating and correlating the strata in which it is found. [1 Mark]
- Must have a wide geographic distribution and a short vertical (geological) time range. [1 Mark]

5. Define free cheek. [K] / [CO2]

- The lateral part of the trilobite cephalon (head) separated from the fixed cheek by the facial suture. [2 Mark]

6. Classify microfossils based on chemical composition. [K] / [CO3]

- **Calcareous** (e.g., Foraminifera), **Siliceous** (e.g., Radiolaria), **Phosphatic** (e.g., Conodonts), or **Organic** (e.g., Pollen). [1 Mark for any two categories, 1 Mark for examples]

7. What does the term 'ancestral traits' mean? [U] / [CO1]

- Characteristics inherited from a distant common ancestor that have remained unchanged through a lineage. [1 Mark]

- **Any Examples:** Body hair in primates is an ancestral trait inherited from a mammalian ancestor. [1 Mark]

8. Define interambulacra. [K] / [CO2]

- In echinoids, these are the five areas between the ambulacral (pore-bearing) rows. [1 Mark]
- They typically consist of larger plates that do not contain tube feet. [1 Mark]

Part B: Short Notes (250 words)

Answer any 3. Each carries 8 marks.

9. Examine the importance of palynology in petroleum exploration. [E] / [CO3]

- **Bio-stratigraphic correlation:** Using pollen/spores to correlate layers across different wells. [3 Marks]
- **Paleo-environmental reconstruction:** Determining if the source rock was marine or terrestrial. [2 Marks]
- **Thermal Maturity:** Analyzing the color of palynomorphs (Thermal Alteration Index) to see if the organic matter reached the "oil window." [3 Marks]

10. Explain tracks, trails, and borings. [U] / [CO1]

- **Tracks:** Individual footprints or impressions left by an organism's feet or appendages. [2 Marks]
- **Trails:** Continuous traces left by an organism crawling or dragging its body (e.g., worm trails). [3 Marks]
- **Borings:** Holes or excavations made by organisms into hard substrates like shells, wood, or rock. [3 Marks]

11. Classify gastropods based on shell characters. [U] / [CO2]

- **Coiling:** Dextral (right-handed) vs. Sinistral (left-handed) **or** planispiral vs. conispiral. [3 Marks]
- **Aperture characteristics:** Presence of a siphonal notch, peristome, or operculum. [2 Marks]
- **Ornamentation:** Ribs, spines, or growth lines on the whorls. [3 Marks]

12. Invertebrate fossils in paleogeographic reconstructions. [E] / [CO4]

- **Continental Positions:** Using fossil distributions (e.g., *Glossopteris* or specific trilobites) to prove former land connections (Gondwana). [4 Marks]
- **Marine Connections:** Identifying shared marine fauna between now-separated basins to indicate ancient seaways or barriers. [4 Marks]

13. Explain conditions of fossilization and note on body fossils. [U] / [CO1]

- **Conditions:** 1. Preservation of original soft part without alteration. 2. Preservation of original hard part without alteration. 3. Preservation of altered hard parts. 4. Preservation of traces or evidence of animal activities. [4 Marks]
- **Body Fossils:** Remains of the actual organism (shells, bones, teeth) preserved via permineralization, replacement, or moulds/casts. [4 Marks]

Part C: Essays (400 words)

Answer any 1. Carries 12 marks.

14. Internal parts of a Cephalopod shell. [U] / [CO2]

- **Septa & Chambers:** Cross-walls (septa) that divide the shell into gas-filled buoyancy chambers (camerae). [3 Marks]
- **Siphuncle:** A tube of living tissue extending through the septa to regulate buoyancy. [3 Marks]
- **Septal Necks:** Extensions of the septa that support the siphuncle at the point of entry. [3 Marks]
- **Connecting Rings:** Segments of the siphuncle between the septal necks. [3 Marks]

15. Palaeontology: Definition, scope, and applications. [U] / [CO1]

- **Definition:** The study of ancient life through fossils. [2 Marks]
- **Scope:** Includes Palaeobotany, Palaeozoology, Micropalaeontology, and Ichnology. [4 Marks]
- **Applications:**
    - **Geological:** Age determination, correlation, and stratigraphy. [3 Marks]
    - **Biological/Industrial:** Understanding evolution, paleoclimate studies, and exploration for fossil fuels (Coal/Oil). [3 Marks]